

Position Generalization in Translated FashionMNIST

Architecture, patch scale, and patch embedding

Final English report

12 fitted models / 24 test evaluations

Abstract. Under one controlled recipe, transfer is asymmetric: model baselines trained on A change by at most 0.73 points at center, while those trained on B lose 57.74-75.58 points at random positions. CNN leads all four settings (35.40% on B -> A). For ViT, patch size 8 leads three settings; Conv2d and Linear projections differ by at most 0.61 points in this single-seed run.

1 Question

A classifier may learn class appearance and object location together. We isolate location by placing each 28 x 28 image either uniformly over all valid integer positions (A) or at the canvas center (B).

Setting	Training	Testing	Purpose
A -> A	random	random	in-distribution reference
B -> B	center	center	fixed-position reference
A -> B	random	center	transfer within A's position support
B -> A	center	random	one-to-many position transfer

2 Controlled protocol

Each of six configurations is fitted once on A and once on B. Validation selects the checkpoint; both test distributions are then measured on the same 10,000 held-out images, producing 12 fits and 24 final evaluations.

Item	Controlled choice
Data	28 x 28 FashionMNIST image on a 64 x 64 black canvas
Split	54,000 train, 6,000 validation; official 10,000-image test set held out
Training	15 epochs; batch 64; AdamW, lr 1e-3, weight decay 1e-4
Selection	best validation checkpoint
Reporting	top-1 test accuracy; elapsed training time; seed 42

Transfer direction. B -> A expands from one observed location to 1,369 valid locations. A -> B evaluates one location already inside the support of A.

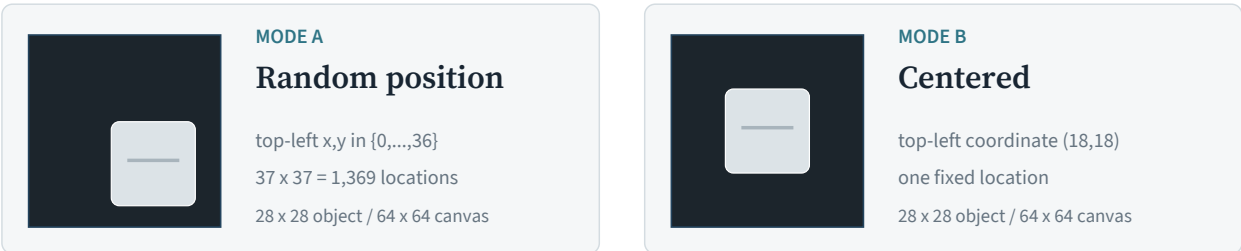


Figure 1. Mode A placements are deterministic per sample and seed; the same test placement seed is used for every configuration.

Architecture and Position Shift

MLP (2 x 128), CNN (5 conv layers), and ViT (d=128, L=4, H=4, p=8)

3 Accuracy

Configuration	Parameters	A -> A	B -> B	A -> B	B -> A
MLP	542,474	71.89	89.57	71.97	13.99
CNN	205,994	92.35	93.14	92.87	35.40
ViT, patch 8	811,402	80.40	88.27	81.13	18.81

Table 1. Top-1 test accuracy (%); teal values mark column leaders.

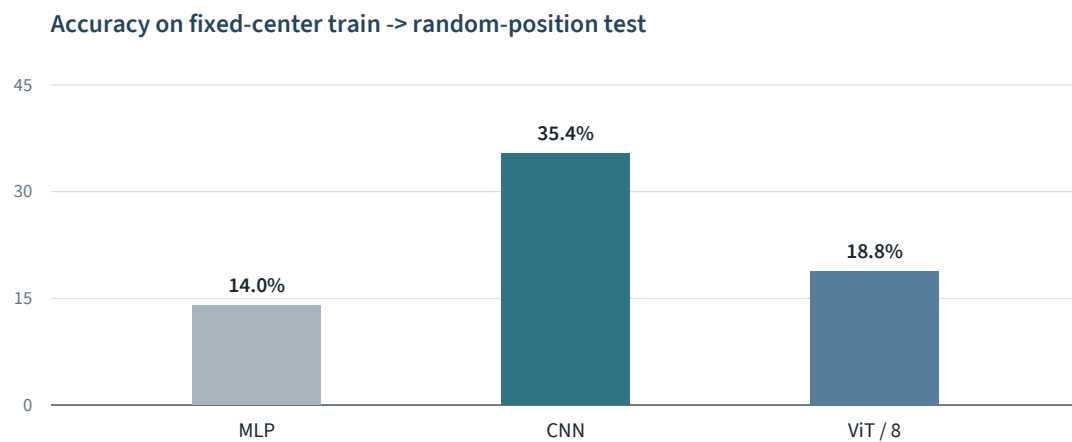


Figure 2. CNN reaches 35.40% on B -> A, 16.59 points above patch-8 ViT.

B -> A LEADER 35.40% CNN	SMALLEST B -> A DROP 57.74 pp CNN, relative to B -> B	A -> B CHANGE 0.73 pp max all three architectures
---	--	--

4 Effect of the position shift

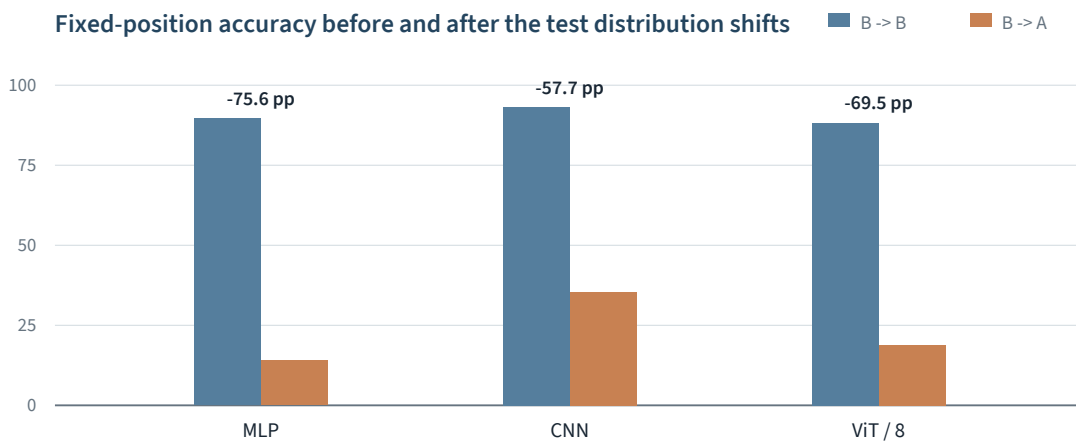


Figure 3. Centered-only training loses 57.74-75.58 points under random-position testing.

Patch Scale and Embedding

ViT body fixed at $d=128$, depth 4, four heads, and FFN width 512

5 Patch scale

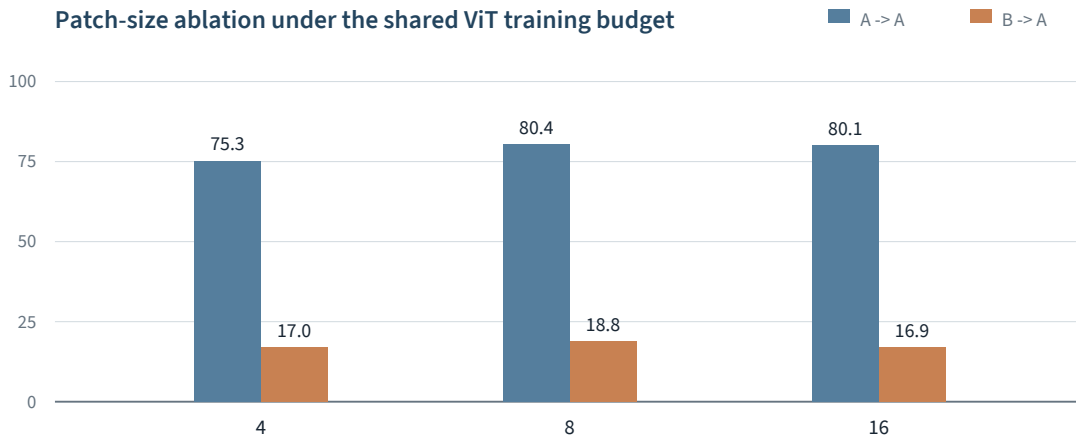


Figure 4. Patch size 8 leads three of four evaluated settings.

Patch	Tokens	A -> A	B -> B	A -> B	B -> A	Time
4	256	75.28	87.22	74.97	17.02	8.93 min
8	64	80.40	88.27	81.13	18.81	7.36 min
16	16	80.07	88.87	80.97	16.91	6.46 min

Table 2. Accuracy (%) and summed training time; teal marks leaders and fastest time.

Result. Patch size 8 leads on A -> A, A -> B, and B -> A. Patch size 4 raises the token count to 256 and is slower without an accuracy gain.

6 Patch embedding

For non-overlapping patches, Conv2d with $\text{kernel} = \text{stride} = p$ and a shared Linear layer on each flattened $p \times p$ patch implement the same affine map when their weights are aligned. Their trained results remain close.

Test	Conv2d	Flatten + Linear	Absolute gap
A -> A	80.07	79.67	0.40
B -> B	88.87	88.80	0.07
A -> B	80.97	80.48	0.49
B -> A	16.91	16.30	0.61

Table 3. Test accuracy (%) for patch size 16. Maximum absolute gap: 0.61 points.

Conclusions and Reproducibility

Concise interpretation and a documented reproduction path

7 **Conclusions**

01	Transfer is strongly asymmetric. A-trained models stay within 0.73 points at center, while B-trained models lose 57.74-75.58 points at random positions. CNN leads all four settings.
02	Patch size 8 leads three of four settings. It gives the best B -> A result without the 256-token cost of patch size 4.
03	Equivalent embeddings stay close in this run. Conv2d and Flatten + Linear differ by at most 0.61 percentage points.

8 **Limitations**

Scope	Interpretation rule
- One training and placement seed; no variance estimate. - Model sizes and optimal hyperparameters are not matched. - Only translation on a black canvas is tested. - Runtime is hardware- and order-specific.	Small differences are described only as numerically close in this run; no significance claim is made. Conclusions apply to the shared protocol rather than all possible training recipes.

9 **Reproducibility**

Check	Recorded value
Code	translated_fashionmnist/experiments
Metrics	results/metrics.csv
Design	12 fits, 24 test evaluations; seed 42; 15 epochs
Dependency lock	requirements-lock.txt
Environment	Python 3.12.13, PyTorch 2.11.0, CUDA 12.8
Hardware	NVIDIA GeForce RTX 5070 Laptop GPU

```
python -m translated_fashionmnist.experiments.compare --groups all --download
```

10 **Implementation map**

Component	Location	Role
Dataset	data.py	load, split, and construct A/B canvases
Models	models.py	MLP, CNN, ViT, patch embeddings
Protocol	experiments/protocol.py	train, validate, and evaluate
Engine	engine.py	shared training and evaluation primitives
Runner	experiments/compare.py	execute all three comparison groups
Outputs	results/	metrics, history, and figures